

The multiplicative forgetful map is not open

Automated mathematics run `fa_banach_001`
Lane 10 — candidate counterexample for expert review

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Status. Candidate full negative answer to Question 6.7 of Goldbring–Sinclair. For every $n \geq 3$, the forgetful map is neither open for the weak topologies nor open for the strong topologies. The proof is complete subject to verification of the standard ternary-envelope lemma isolated below.

1 Source question

Goldbring and Sinclair define $\mathcal{X}_n$ to retain the quantifier-free multiplicative structure of an n -dimensional unital operator subspace, while $\mathcal{OS}_n$ remembers only its operator-space structure. They prove that the forgetful map

$$G : \mathcal{X}_n \longrightarrow \mathcal{OS}_n$$

is weak–weak and strong–strong continuous. Question 6.7 on printed page 17 asks whether G is open for every n [1].

In light of Theorem 6.6 and the fact that $\mathcal{E}_n$ is weakly dense in $\mathcal{OS}_n$ for each n , a positive answer to the following question would imply that KEP holds:

Question 6.7. *Is the forgetful map $G : \mathcal{X}_n \rightarrow \mathcal{OS}_n$ open for each n ?*

7. THE LOCAL LIFTING PROPERTY

Figure 1: Question 6.7 in the source paper. The crop is rendered from the local copy `source_paper.pdf`.

The source defines the $\mathcal{X}_n$ weak and strong metrics through quantities $\gamma_m(\phi)$ that compare the norms of all noncommutative $*$ -polynomials of degree at most m under a unital linear bijection ϕ . We denote the resulting metrics by γ_w and γ_s . The weak metric is a weighted sum of the degree- m discrepancies; the strong metric takes their supremum.

2 Proof intuition

Start with the n -point simplex, whose affine function system is the whole algebra $\mathbb{C}^n$. Add one vertex just beyond a face but let it tend to the midpoint of an edge. Affine matrix-valued functions cannot distinguish this new vertex in the limit, so after forgetting multiplication the resulting spaces converge even in completely bounded distance.

Multiplication does distinguish it. The coordinate functions on $\mathbb{C}^n$ are orthogonal projections. If representatives of the perturbed operator spaces converged multiplicatively, their coordinate

images would have all cross ternary products xx^*y tending to zero. The ternary envelope makes this conclusion independent of the chosen C^* -cover. In the limit, Banach–Stone rigidity forces the coordinates to be a phased permutation. At the extra vertex, two coordinates then have modulus $1/2$, leaving a cross ternary product of modulus $1/8$.

3 The simplex family

Fix $n \geq 3$ and put $E_0 = \mathbb{C}^n$ with its canonical C^* -algebra operator space structure. For $0 < \varepsilon < 1$, set

$$a_\varepsilon := \frac{1 + \varepsilon}{2}, \quad L_\varepsilon(z) := a_\varepsilon z_1 + a_\varepsilon z_2 - \varepsilon z_3,$$

and define

$$J_\varepsilon : \mathbb{C}^n \longrightarrow \mathbb{C}^{n+1}, \quad J_\varepsilon(z) = (z_1, \dots, z_n, L_\varepsilon(z)), \quad S_\varepsilon := J_\varepsilon(\mathbb{C}^n).$$

Since $2a_\varepsilon - \varepsilon = 1$, S_ε contains the unit. It is self-adjoint and separates the $n + 1$ coordinates, hence $C^*(S_\varepsilon) = \mathbb{C}^{n+1}$.

Geometrically, S_ε is the space of affine functions on

$$V_\varepsilon = \{e_1, \dots, e_n, w_\varepsilon\}, \quad w_\varepsilon = a_\varepsilon e_1 + a_\varepsilon e_2 - \varepsilon e_3,$$

inside the real affine hyperplane $\sum_i t_i = 1$. All $n + 1$ points are exposed vertices: e_1, e_2 maximize their own coordinates, e_3 maximizes the third coordinate, e_i for $i \geq 4$ maximizes the i th coordinate, and w_ε uniquely minimizes the third coordinate. Thus every point is a peak point for S_ε . Consequently its Shilov boundary is all of V_ε , and its C^* - and ternary envelopes are $\mathbb{C}^{n+1}$.

Lemma 1 (Uniform matrix-level convergence). *For every matrix level k ,*

$$\|J_\varepsilon^{(k)}\| \leq 1 + 2\varepsilon, \quad \|(J_\varepsilon^{-1})^{(k)}\| \leq 1.$$

Therefore $G(S_\varepsilon) \rightarrow G(E_0)$ in the strong topology of $\mathcal{OS}_n$, and hence also in the weak topology.

Proof. For $Z = (Z_1, \dots, Z_n) \in M_k(\mathbb{C}^n)$, put $M = \max_i \|Z_i\|$. Then

$$\|J_\varepsilon^{(k)}(Z)\| = \max \{M, \|a_\varepsilon Z_1 + a_\varepsilon Z_2 - \varepsilon Z_3\|\}.$$

The second entry is at most $(2a_\varepsilon + \varepsilon)M = (1 + 2\varepsilon)M$, while the first n coordinates show that $\|J_\varepsilon^{(k)}(Z)\| \geq M$. $\square$

4 The cover-independent obstruction

We use the following standard consequence of Hamana’s ternary-envelope universal property [2]. It is stated explicitly to isolate the only operator-space structure theorem used in the proof.

Lemma 2 (Ternary-envelope contraction). *Let F be a unital operator subspace of a C^* -algebra and let $u : F \rightarrow S_\varepsilon$ be a surjective complete isometry; u need not be unital. Then for all $x, y, z \in F$,*

$$\|u(x)u(y)^*u(z)\|_{\mathbb{C}^{n+1}} \leq \|xy^*z\|_{C^*(F)}.$$

Proof. Because F contains the unit, the TRO generated by F inside $C^*(F)$ is all of $C^*(F)$. The universal property gives a surjective ternary morphism from this TRO cover onto the ternary envelope $T(F)$. The complete isometry u induces a ternary isomorphism $T(F) \cong T(S_\varepsilon) = \mathbb{C}^{n+1}$ carrying the canonical image of x to $u(x)$. Ternary morphisms preserve xy^*z and are contractive, which gives the inequality. $\square$

Lemma 3 (No multiplicative lifts). *Suppose $\varepsilon_j \downarrow 0$ and $F_j \in \mathcal{X}_n$ satisfy $G(F_j) = G(S_{\varepsilon_j})$. Then neither $\gamma_w(F_j, E_0) \rightarrow 0$ nor $\gamma_s(F_j, E_0) \rightarrow 0$ is possible.*

Proof. Either convergence would provide unital linear bijections $\phi_j : E_0 \rightarrow F_j$ for which

$$\gamma_1(\phi_j) \rightarrow 0, \quad \gamma_3(\phi_j) \rightarrow 0.$$

Choose surjective complete isometries $u_j : F_j \rightarrow S_{\varepsilon_j}$, which need not be unital, and define

$$A_j := J_{\varepsilon_j}^{-1} u_j \phi_j : \mathbb{C}^n \rightarrow \mathbb{C}^n.$$

Testing γ_1 on the polynomial $p(X) = X$ gives, uniformly in z ,

$$(1 - o(1))\|z\|_{\infty} \leq \|u_j \phi_j(z)\| \leq (1 + o(1))\|z\|_{\infty}.$$

Together with Lemma 1, this yields

$$\frac{1 - o(1)}{1 + 2\varepsilon_j} \|z\|_{\infty} \leq \|A_j z\|_{\infty} \leq (1 + o(1))\|z\|_{\infty}.$$

After taking a subsequence, the finite-dimensional maps A_j converge to a surjective linear isometry $A : \ell_{\infty}^n \rightarrow \ell_{\infty}^n$. By the finite-dimensional Banach–Stone theorem, there are a permutation σ and unimodular scalars λ_r such that

$$Ae_r = \lambda_r e_{\sigma(r)} \quad (1 \leq r \leq n).$$

Choose distinct r, s with $\sigma(r) = 1$ and $\sigma(s) = 2$, and put $f_{j,t} = u_j \phi_j(e_t) = J_{\varepsilon_j}(A_j e_t)$. At the extra coordinate,

$$f_{j,t}(n+1) = a_{\varepsilon_j}(A_j e_t)_1 + a_{\varepsilon_j}(A_j e_t)_2 - \varepsilon_j(A_j e_t)_3.$$

It follows that

$$f_{j,r}(n+1) \rightarrow \frac{\lambda_r}{2}, \quad f_{j,s}(n+1) \rightarrow \frac{\lambda_s}{2},$$

and hence

$$\liminf_j \|f_{j,r} f_{j,r}^* f_{j,s}\|_{\mathbb{C}^{n+1}} \geq \frac{1}{8}. \quad (1)$$

On the other hand, $e_r e_r^* e_s = 0$ in $E_0 = \mathbb{C}^n$. Testing $\gamma_3(\phi_j)$ on the degree-three polynomial $p(X, Y) = XX^*Y$ gives

$$\|\phi_j(e_r) \phi_j(e_r)^* \phi_j(e_s)\|_{C^*(F_j)} \leq \gamma_3(\phi_j) \rightarrow 0.$$

Lemma 2 contradicts (1). □

5 Negative answer to Question 6.7

Theorem 4. *For every $n \geq 3$, the forgetful map $G : \mathcal{X}_n \rightarrow \mathcal{OS}_n$ is not weak-open and is not strong-open. In particular, Question 6.7 has a negative answer under either topology introduced in the source paper.*

Proof. We give the weak argument; the strong argument is identical. Suppose G were weak-open at E_0 . For each k , let

$$U_k = \{F \in \mathcal{X}_n : \gamma_w(F, E_0) < 1/k\}.$$

Then $G(U_k)$ is a neighborhood of $G(E_0)$. By Lemma 1, $G(S_{1/m}) \rightarrow G(E_0)$, so one may choose $m_k \geq k$ and $F_k \in U_k$ such that

$$G(F_k) = G(S_{1/m_k}).$$

Thus $1/m_k \rightarrow 0$ and $\gamma_w(F_k, E_0) \rightarrow 0$, contradicting Lemma 3. Replacing γ_w by γ_s proves failure of strong openness. □

6 Verification, scope, and novelty bounds

The argument is non-computational. The following points were checked independently in the proof:

1. the coefficient sum in L_ε is 1, so the construction is unital;
2. the matrix-level distortion is bounded by $1 + 2\varepsilon$ uniformly in k ;
3. all $n + 1$ evaluation points are exposed, so no coordinate disappears in the Shilov/ternary envelope;
4. arbitrary, possibly nonunital complete isometries are handled by the ternary envelope rather than by an unjustified multiplicative extension;
5. the extra-coordinate obstruction is exactly $(1/2)^2(1/2) = 1/8$;
6. nested open balls convert openness into the forbidden sequence of multiplicative lifts.

The construction covers every $n \geq 3$. It does not determine openness for $n = 1, 2$, which is not needed to refute the statement “open for each n .”

A bounded novelty search on 9 August 2026 covered the run’s registry, solution, attempt, and proof-gap indexes; the exact wording of Question 6.7; arXiv:1501.06395 and the source title; later papers citing the source; and web queries combining “forgetful map,” $\mathcal{X}_n$, $\mathcal{OS}_n$, and “Kirchberg embedding problem.” No later explicit resolution or simplex construction was found. This is not an exhaustive MathSciNet or zbMATH priority search.

Human-review recommendation. High priority. Verify Lemma 2 against Hamana’s universal property, then check the compactness passage to the phased permutation and the weak/strong open-map quantifiers. The remaining estimates are elementary.

References

- [1] I. Goldbring and T. Sinclair, *Omitting Types in Operator Systems*, Indiana Univ. Math. J. 66 (2017), no. 3, 821–844; arXiv:1501.06395; DOI: [10.1512/iumj.2017.66.6019](https://doi.org/10.1512/iumj.2017.66.6019).
- [2] M. Hamana, *Triple Envelopes and Šilov Boundaries of Operator Spaces*, Math. J. Toyama Univ. 22 (1999), 77–93; DOI: [10.15099/00003027](https://doi.org/10.15099/00003027).